

1. Assume that a memory access takes 20 ns if it results in a cache hit, and 50 ns if it results in a cache miss. The current state of the direct-mapped cache is shown below:

Cache Table:

Index	Valid	Tag	Data
000	1	100	Data[100000]
001	0/1	011	Data[011 001]
010	0/1	100	Data[100 010]
011	0/1	100	Data[100 011]
100	1	100 110 011 11001	Data[100100] Data[110100] Data[011100] Data[111100]
101	0		
110	1	011	Data[011110]
111	1	111	Data[111111]

Data[001 100]

The size of a block is 8 bits. Using the table below, calculate the total access time for the given memory sequence using a direct-mapped cache. Indicate hit/miss and access time for each entry. Write your final answers in the table below.

Access Table:

Memory Address	Binary Address	Hit/Miss	Access Time
35	100 011	miss	50
52	110 100	miss	50
28	011 100	miss	50
60	111 100	miss	50
32	100 000	Hit	20
12	001 100	miss	50

25	011 001	Miss	50
34	100 010	Miss	50
Total Access Time			370 ns

a) Complete the cache table and access table simulation in the given table above. [8+2]

b) Find the AMAT. [1]

$$370 / 8 = 46.25 \text{ ns}$$

Q2: Suppose a cache has 16 blocks. Each block can store 16 bits of data.

a) What is the size of the cache? [1]

Answer: $16 \times 16 = 256 \text{ bits} = 32 \text{ bytes}$

b) Using the mentioned cache system, for a memory address 1010100101, Find the index, tag and offset bits. [3]

Answer:

$$\text{offset} = \log_2(2) = 1 \text{ bit}$$

$$\text{Index} = \log_2(16) = 4 \text{ bits}$$

$$\text{Tag} = 10 - 4 - 1 = 5 \text{ bits}$$

Index bits = 0010

Offset bits = 1

Tag bits = 10101

c) If the access results in a cache hit, draw a flowchart of the write back process. [5]

CPU $\xrightarrow{\text{mem acc.}}$ Cache $\xrightarrow{\text{hit}}$ write on Cache + Dirty block $\xrightarrow{\text{mark}}$

On replacing dirty block $\Rightarrow$ write on RAM first.

1. Assume that a memory access takes 20 ns if it results in a cache hit, and 50 ns if it results in a cache miss. The current state of the direct-mapped cache is shown below:

Cache Table:

Index	Valid	Tag	Data
000	1	100 010	Data[100000] Data[111 000] Data[010 000]
001	0		
010	0 1	0 11	Data[0 11 010]
011	0		
100	1	100	Data[100100]
101	0 1	011 100	Data[011 101] Data[100 101]
110	1	011	Data[011110]
111	1	111	Data[111111]

The size of a block is 16 bits. Using the table below, calculate the total access time for the given memory sequence using a direct-mapped cache. Indicate hit/miss and access time for each entry. Write your final answers in the table below.

Access Table:

Memory Address	Binary Address	Hit/Miss	Access Time
36	100 100	Hit	20
56	111 000	Miss	50
26	011 010	Miss	50
63	111 111	Hit	20
30	011 110	Hit	20
16	010 000	Miss	50

29	011 101	miss	50
37	100 101	miss	50
Total Access Time			310

a) Complete the cache table and access table simulation in the given table above. [8+2]

b) Find the AMAT. [1]

$$310/8 = 38.75 \text{ ns}$$

Q2: Suppose a cache has 16 blocks. Each block can store 32 bits of data.

a) What is the size of the cache? [1]

Answer: $16 \times 32 = 512 \text{ bits} = 64 \text{ bytes}$.

b) Using the mentioned cache system, for a memory address 1011100001, Find the index, tag and offset bits. [3]

Answer:

$$\text{offset bitsize} = \log_2(\# \text{ bytes in block}) = \log_2(4) = 2$$

$$\text{Index} = \log_2(\# \text{ block in cache}) = \log_2(16) = 4$$

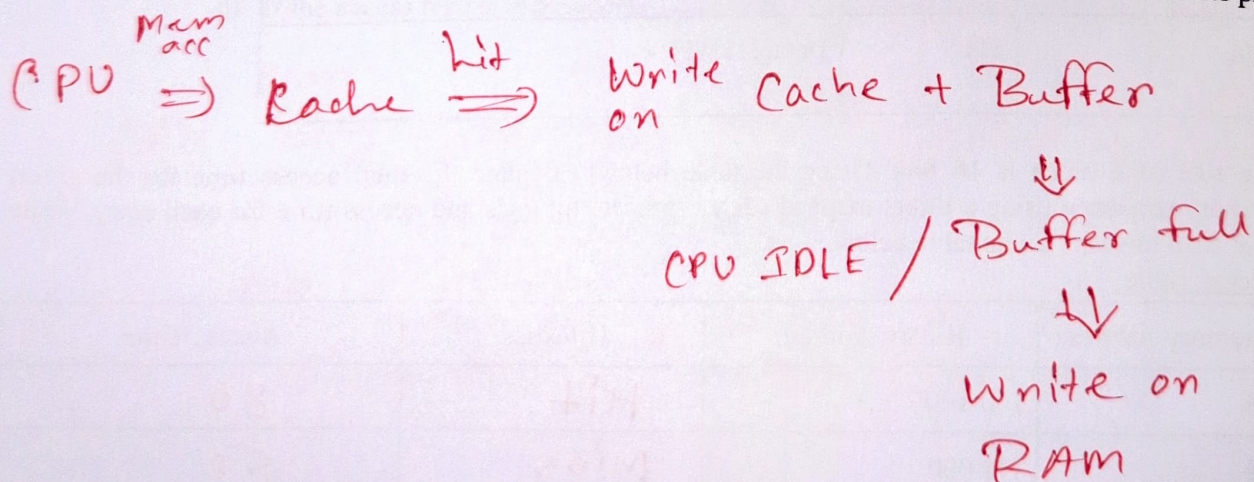
$$\text{Tag} = 10 - 4 - 2 = 4$$

Index bits = 1000

Offset bits = 001

Tag bits = 1011

c) If the access results in a cache hit, draw a flowchart of the write through with the buffer process. [5]



Assume that a memory access takes 20 ns if it results in a cache hit, and 50 ns if it results in a cache miss. The current state of the direct-mapped cache is shown below:

Cache Table:

Index	Valid	Tag	Data
000	1	100 111 010	Data[100 000] Data[111 000] Data[010 000]
001	0		
010	0 1	011	Data[011 010]
011	0		
100	1	100	Data[100 100]
101	0 1	011 100	Data[011 101] Data[100 101]
110	1	011	Data[011 110]
111	1	111	Data[111 111]

The size of a block is 8 bits. Using the table below, calculate the total access time for the given memory sequence using a direct-mapped cache. Indicate hit/miss and access time for each entry. Write your final answers in the table below.

Access Table:

Memory Address	Binary Address	Hit/Miss	Access Time
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63	111 111	Hit	20
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16	010 000	Miss	50

29	011 101	Miss	50
37	100 101	Miss	50
Total Access Time			310ns

a) Complete the cache table and access table simulation in the given table above. [8+2]

b) Find the AMAT. [1]

$$310/8 = 38.75ns$$

Q2: Suppose a cache has 16 blocks. Each block can store 48 bits of data.

a) What is the size of the cache? [1]

$$\text{Answer: } 16 \times 48 = 768 \text{bits}$$

b) Using the mentioned cache system, for a memory address 1011100001, Find the index, tag and offset bits. [3]

Answer:

$$\text{Index bitsize} = \log_2(\# \text{blocks in cache}) = 4 ; \text{Offset bitsize} = \log_2(\# \text{bytes in block}) = 2.58 \approx 3$$

$$\text{Tag} = 10 - 4 - 3 = 3$$

Index bits = 1100

Offset bits = 001

Tag bits = 101

c) If the access results in a cache miss, draw a flowchart of the write allocation process. [5]

CPU mem access => Cache Miss => Copies the block from mem to cache (allocation on miss) => Writes in cache through cache hit processes.

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010	0 1	100	Data[100 010]
011	0 1	100	Data[100 011]
100	1	100 100 011 1100	Data[100100] Data[110100] Data[111100] Data[111100] Data[001100]
101	0		
110	1	011	Data[011110]
111	1	111	Data[111111]

The size of a block is 16 bits. Using the table below, calculate the total access time for the given memory sequence using a direct-mapped cache. Indicate hit/miss and access time for each entry. Write your final answers in the table below.

Access Table:

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34	100 010	miss	50
Total Access Time			370ns

a) Complete the cache table and access table simulation in the given table above. [8+2]

b) Find the AMAT. [1]

$$370 / 8 = 46.25 \text{ ns}$$

Q2: Suppose a cache has 8 blocks. Each block can store 64 bits of data.

a) What is the size of the cache? [1]

Answer: $8 \times 64 = 512 \text{ bits} = 64 \text{ bytes}$

b) Using the mentioned cache system, for a memory address 1010100101, Find the index, tag and offset bits. [3]

Answer:

$$\text{offset bitsize} = \log_2(\# \text{ bytes in block}) = \log_2(8) = 3$$

$$\text{Index } u = \log_2(\# \text{ blocks in cache}) = \log_2(8) = 3$$

$$\text{Tag } u = 10 - 3 - 3 = 4$$

Index bits = 100

Offset bits = 101

Tag bits = 1010

c) If the access results in a cache miss, draw a flowchart of the write around process. [5]

CPU $\xRightarrow{\text{mem access}}$ Cache $\xRightarrow{\text{miss}}$ Halt

CPU $\Rightarrow$ RAM
Write Data